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Project Plan

for

Project F26-4.3:
Resonant Tuning Fork Thermometry

Client: Maybell Quantum

Team Members:

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1. Project Plan

1.1. Project Name
F26-4.3 Resonant Tuning Fork Thermometry

1.2. Client
Maybell Quantum
Kyle Thompson - kyle@maybellquantum.com

1.3. High-level Objective
This project aims to measure dilution fridge temperatures using resonant tuning fork thermometry.

1.4. Organization
Technical Advisors:

- Andrew Woods - awoods@magnet.fsu.edu
- David Kendellen - david@maybellquantum.com

Project Advisor:

- Eric Brighton - eric.brighton@mines.edu

Student team:

- Evan Christie – christie@mines.edu – Communication Lead
- Enzo Guglielmo – enzo_guglielmo@mines.edu
- Nathan Johnston – nathan_johnston@mines.edu – CFO
- Ryan Montoya – ryan_montoya@mines.edu
- Zurin Trzaska – zurin_trzaska@mines.edu – Scrum Master
- Anastasiia Usmanova – anastasiia_usmanova@mines.edu

2. Background & Problem Identification

2.1. Scope of Work

The current prototype is not as accurate as the client would like it to be, and it is not scalable for manufacturing. The significance of developing the manufacturable device is to reduce the cost, as well as decrease the chances of expensive Helium (He) leakage. The current readout scheme is reflective of a laboratory setup rather than a stand-alone product.

2.2. Background Research

Existing research utilizing a quartz tuning fork as a cryogenic thermometer and viscometer exists, illustrating that the development of a full manufacturable solution is the next viable step in the technological development rather than it being an open scientific question.

2.2.1. "Quartz Tuning Fork: Thermometer, Pressure- and Viscometer for Helium Liquids" [1]

Researchers developed a foundational tuning-fork model to serve as a low-cost, field-insensitive probe usable as a pressure gauge, viscometer, and thermometer. This was done by treating each tine of the fork as a damped harmonic oscillator. From that, they were able to derive mathematical expressions to connect the resonant frequency and resonance width to the density and viscosity of the surrounding liquid. The paper also illustrates some weaknesses of the technology, including that vacuum quality factors and intrinsic dissipation can vary by more than an order of magnitude between nominally identical, commercially sourced forks. Furthermore, it was found that hydrodynamic fit parameters can differ significantly fork-to-fork, even for forks of identical geometry. The model in the study also breaks down once the viscous penetration depth becomes comparable to the fork's inter-leg spacing, which occurs near the superfluid transition temperature in ^{3}He and at the lowest temperatures generally, the model can't address it. The overall conclusion was that forks require individual preselection and calibration rather than a universal factory calibration, which is a significant practical limitation for any product built on the technology.

2.2.2. "Developing compact tuning fork thermometers for sub-mk temperatures and high magnetic fields" [2]

A National High Magnetic Field Laboratory and University of Florida study pushed tuning-fork thermometry to temperatures down to 1mK and magnetic fields up to 14.5 T. The researchers showed that their hydrodynamic model fails around 1-10 mK in normal ^{3}He liquid due to the viscosity being so high that the flow can no longer follow the tines. This means that when the viscosity is high, the fluid can no longer keep up with the fork's motion and create a slip. To address this, they introduced a slip-corrected hydrodynamic function to be able to mathematically describe that fluid drag on the fork that builds in this slip effect. Furthermore, the study found that the tuning-fork constant is field independent within 0.6% up to 14.5 T. Building upon that idea, they presented a miniaturized thermometer package that consisted of a tuning fork and a silver sinter heat exchanger that was sealed inside a silver tube. This was designed to thermally link the fork with solid samples via liquid ^{3}He. Furthermore, utilizing the specific heat capacities of silver and ^{3}He, they estimated the total package heat capacity of only around 4 μJ/K with an average thermal response of around 50 seconds. This model is the

closest existing design to a self-contained, packaged tuning fork thermometer, compared to a bare fork instrumented directly in a cryostat.

2.2.3. "Thermometry in Normal Liquid 3He Using a Quartz Tuning Fork Viscometer" [3]

This study fabricated a tuning fork thermometer that covers a range of 6 mK to 1.8 K in normal liquid ^{3}He at saturated vapor pressure. The way that this was possible was by cross calibrating the fork against a ruthenium-oxide resistor above 40 mK and a vibrating wire resonator below it. Rather than utilizing multiple hydrodynamic parameters, the fork was fixed with a single geometrical constant C using one reference point against prior wire viscosity data. The resulting model reproduces the full temperature dependent viscosity of ^{3}He that was very accurate with independent measurements. The limitation of this study is that due to their calibration technique, it depends on an independent reference thermometer to fix C in the first place. So, it simplifies the fitting, but it doesn't eliminate the need to crosscalibration against another sensor.

Taken together, these three sources show a clear progression of tuning fork thermometry technology. Blaauwgeers et al. established the viability of the technology but left fork-to-fork variability and high viscosity breakdown as open issues. Woods et al. addressed the high-viscosity issues with a slip-corrected model and a prototype but had to add an empirical fitting parameter and had to create a low-temperature and high-resilient enclosure. Bradley et al. showed the model could be reduced to a single calibration parameter, but only when the viscous penetration depth stays well below the fork prong width and still relying on an independent reference thermometer to fix the parameter. The remaining gap that is being targeted is the intersection of these three studies. The ultimate goal is a compact, low-thermal-mass tuning fork package that is validated for fast thermal measurements on small samples under these combined ultra-low-temperature and high-field conditions.

2.3. Goal

The project goal is to develop a manufacturable design for a self-contained thermometry device utilizing resonant tuning forks in liquid ^{3}He to accurately measure within 1 mK the temperature of a cooling plate in a dilution refrigerator down to temperatures of 7 mK at a sampling interval of 30 seconds.

2.4. Stakeholders

Table 1 represents the stakeholders and the involvement the team anticipates.

Table 1: Stakeholder's Involvement

Stakeholder	**Involvement**
Maybell Quantum	Dictates design requirements, manufacturability requirements, and budget
Maybell Quantum's customers (quantum computing technology companies)	Dictates quality and longevity requirements
Seiko	Supply tuning forks
Other materials suppliers (Unknown)	Supply components

2.5. Relevant Engineering Standards

PLTS-2000 *Provisional Low Temperature Scale of 2000* – Defines temperatures between 0.9mK and 1.0K based on ^{3}He melting pressure curve. Used to validate very low temperature measurements. [4]

ITS-90 *International Temperature Scale of 1990* – Defines temperatures between 0.65K and 1358K based on the triple point of water. Used to validate room to low temperature measurements. [5]

ASTM E595 *Standard Test Method for Total Mass Loss (TML) and Collected Volatile Condensable Materials (CVCM) from Outgassing in a Vacuum Environment* – Quantifies a method of measuring outgassing of materials to be used in vacuum environments of $5 \times 10^{-5}\ torr$ and temperatures up to $125℃$. Used to inform and validate material selections in design. [6]

3. Statement of Work & Requirements

3.1. Customer Needs

Table 2 summarizes the client's design requirements. The table is split into two sections: design requirements (the items that are non-negotiable, i.e. "client needs") and design stretch goals (additional items that would improve our design but are negotiable, i.e. "client wants"). Design stretch goals that are extensions/related to a design requirement have been placed in the same row. The Helium Thermometry Cell (HTC) is the component that contains the helium mixture and houses the tuning fork. The readout device is responsible for driving, analyzing, and displaying information sourced from the HTC.

Table 2: Customer needs

Design Requirements	**Design Stretch Goals**
The readout device shall measure temperature within ±1mK while operating at the lower limit.	The readout device should infer mixture composition.
The readout device shall measure the temperature down to the lower limit of T = 7 mK.	The readout device should reach measurement temperatures below 7 mK.
The HTC shall store a ^{3}He/^{4}He mixture with no detectable leakage.	
The team shall demonstrate the manufacturability of the HTC and readout device within the unit cost budget, and lead time.	The team should exhibit manufacturability through a pilot batch and manufacturing capability analysis.
The RD's polling rate shall be less than 30 seconds.	The readout device should achieve a measurement polling rate of less than 10 seconds.
The readout device shall be integrated into the secondary equipment rack, using available power supplies and physical rack space.	
The readout device shall display driving voltage, response current and frequency, and calculated temperature.	The readout device should interface with existing Maybell software system architecture.
The team shall provide a calibration method for fork-to-fork manufacturing variance.	
The HTC shall maintain operational reliability in low temperature and high magnetic (up to 14T) environments.	
The HTC to readout device interconnection shall minimize induced noise/EMI within the chamber.	
The device shall minimize heat generation within the chamber through wiring and non-magnetocaloric materials.	

3.2. Design Requirements

Table 3 represents the design requirements. This table outlines the criteria for meeting design requirements and will track the design progress towards meeting these requirements. The customer needs are already expressed in a technical manner, so minimal interpretation is needed to create measurable design requirements.

Table 3: Design Requirements (see appendices for enlarged version)

Parent ID	Requirement ID	Requirement Type	Status	Requirement	Verification Type	Rationale/Justification	Notes
SubSys1	1.1	Performance	Draft	Measure temperature within +/- 1mK at low temperatures.	Test	Precision specified by client.	
SubSys1	1.2	Performance	Draft	Measurement Lower Limit: Measure the temperature down to 7 mK	Test	Limit specified by client.	
SubSys3	3.1	Functional	Draft	Contain $^3He/^4He$ mixture with no detectable leakage	Test	Any He leak will compromise function of device.	
SubSys3	3.2	Functional	Draft	Demonstrate manufacturability of the design within unit cost budget, and lead time	Demonstration	Client wants device to be manufacturable.	
SubSys2	2.1	Performance	Draft	Measurement polling rate of less than 30s	Test	Polling rate specified by client.	
SubSys2	2.2	Functional	Draft	Device integrated on secondary equipment rack, using available power supplies and physical rack space	Inspection	Installation parameters specified by client.	
SubSys2	2.3	Functional	Draft	Readout info: Display on rack with voltage, current, frequency, and calculated temperature	Inspection	Temperature display required by client, other values used for debugging and performance monitoring.	
SubSys1	1.3	Performance	Draft	Develop calibration scheme for fork-to-fork manufacturing variance	Demonstration	Contributes to manufacturability and consistent function of device across batches.	
SubSys1, SubSys3	1.4, 3.3	Performance	Draft	Operational reliability in low temp and high magnetic fields up to 14T	Test	Operate in conditions and environment specified by client.	
SubSys1	1.5	Performance	Draft	Minimize noise and EMI generation within the chamber	Analysis	Cannot interfere with performance of fridge or devices within fridge.	
SubSys1, SubSys4	1.6, 4.1	Performance	Draft	Minimize heat generation within chamber through wiring and non-magnetocaloric materials	Analysis	Cannot interfere with performance of fridge or devices within fridge.	

3.3. Boundaries

This project is shaped by a set of constraints, operating limitations, and environmental factors that define the space in which the tuning fork design must succeed. Constraints that must be considered are the requirements around size, manufacturability, cost, and electromagnetic interference that the fork might face. This means that it must be compact enough to fit within the intended deployment area and producible with available fabrication budget.

The project also operates within specific limitations including the budget provided, the timeline, and the required temperature range over which the device must reliably measure. These limitations set the practical boundaries of which, the final design must function.

Environmentally, the project is bound by how the final product will be deployed in the real world. The tuning fork is intended for use within Maybell commercial refrigerators and people who interact with the system. For this reason, handling, installation, and operation must be accounted for in the design to ensure safety and usability.

Finally, the project is limited by concrete budget and resource constraints. The total budget for this project is $3,500 and the overall timeline runs from September 2026 through April 2027. The team is not expected to work during university breaks or federal holidays, which slightly narrows the available working windows within the total timeline.

4. Methods of Approach

4.1. Design process

Development of a manufacturable tuning fork thermometry instrument will proceed with a cyclical engineering process, with focus on consistent client communication, Internal team design analysis, and on-site testing (materials, conditions, etc.).

A detailed work breakdown structure (WBS) is provided in Appendix c.

4.2. Subsystems

1.1 Sensing
- 1.1.1 Components: Tuning fork, soldering and wiring, Electromagnetic Compatibility design requirements
- 1.1.2 Inputs: power, ^{3}He mixture
- 1.1.3 Outputs: temperature, breakdown of the He mixture
- 1.1.4 Interfaces: Readout, Housing and Hardware Integration

1.2 Readout
- 1.2.1 Signal interpretation, display/UI, signal filtering
- 1.2.2 Inputs: Electrical Signal
- 1.2.3 Outputs: Temperature estimation
- 1.2.4 Interfaces: Digital readout

1.3 Housing and Hardware Integration
- 1.3.1 Fluid housing, electronics housing and interface, integration into refrigerator
- 1.3.2 Inputs: refrigerator geometry, sensor and electrical component geometry, refrigerator environment conditions
- 1.3.3 Interfaces: Sensing, Thermal/Fluid

1.4 Thermal/Fluid
- 1.4.1 He containment and pressure management, thermal conductivity analysis and design, insertion of He into system during manufacturing
- 1.4.2 Inputs: refrigerator environment conditions, ^{3}He properties and purity
- 1.4.3 Outputs: housing geometry, housing materials, electrical materials
- 1.4.4 Interfaces: Housing and Hardware Integration

4.3. Schedule for Deliverables

Table 4 represents the schedule for deliverables, highlighting the responsibilities party, tentative delivery date and showing whether that task is a part of the critical path.

Table 4: Schedule for deliverables

Deliverable	**Done when**	**Responsible party**	**Duration**	**Delivery date**	**Delay mitigation**
Requirements	Goals and constraints approved	Team	4 weeks	Oct-26	Confirm decisions with client early
Concept design	Design approach selected	Team	5 weeks	Nov-26	Keep a backup concept
Preliminary Design Review	Feedback recorded	Team	1 day	Nov-26	Schedule review early
Detailed design	Drawings and calculations complete	Team	8 weeks	Jan-27	Check work as it is completed
Critical Design Report + Calculation Package	Submitted	Team	1 week	Spring week 2	Draft during design
Critical Design Review	Design reviewed	Team	1 day	Spring week 3	Book reviewers early
Prototype build	Prototype assembled	Team	6 weeks	Mar-27	Reserve fabrication time
Testing	Results compared with goals	Team	4 weeks	Apr-27	Reserve test time; allow retesting
Final Design Report	Submitted	Team	1 week	Spring week 13	Write sections throughout project
Final Design Review	Results reviewed	Team	1 day	Spring week 14	Schedule before Showcase
Design Showcase	Project presented	Team	1 day	Spring week 15; date TBD	Prepare backup demonstration
Revised Final Design Report	Feedback incorporated	Team	1 week	Spring week 16	Reserve time for revisions

5. Project Execution

5.1. Performance Criteria & Verification Plan

The team will need to prove the tuning fork's capabilities to measure temperature first on the bench at room temperature, and then in a lab environment and down to 7 mK level temperatures. The team will need to show the thermometer's ability to capture the temperature readings no longer than 30 seconds between readings, along with all other technical constraints detailed in the design requirements. The team will need to show that the thermometers can be manufactured and reproduced for commercial usage.

The Requirement Verification Matrix (Table 5) outlines the testing procedures to be used in testing the prototype.

Table 5: Performance criteria and verification plan (see appendices for enlarged version)

Requirement ID	Requirement Type	Performance Parameter	Verification Type	Measurement Method	Target/ Threshold (Acceptance Criteria)	Verification Procedure	Status
1.1	Performance	Difference between device measured temp and calibrated instrument measured temp	Test	Calculation	1mK in the low temperature range (<1K)	Compare device's temperature measurement to existing thermometry solution's temperature measurement.	Unverified
1.2	Performance	Difference between device measured temp and calibrated instrument measured temp	Test	Calculation	7mK with precision described by requirement 1.1	Compare device's temperature measurement to existing thermometry solution's temperature measurement.	Unverified
3.1	Functional	ppm/s He detected	Test	Measurement	Below detection limit of He leak check device	Use vacuum/helium leak detection device.	Unverified
3.2	Functional	Adherence to manufacturing process across repeat assemblies of device	Demonstration	Analysis	Six Sigma manufacturing principles	Analyze variance in device performance throughout production of pilot batch	Unverified
2.1	Performance	Temperature readings per second	Test	Measurement	1/30 temperature readings per second (or 30 seconds per temperature reading)	Externally count time between temperature reading updates.	Unverified
2.2	Functional	Completeness of implementation into existing hardware	Inspection	Observation	100%	Observe implementation of device power draw and support systems.	Unverified
2.3	Functional	Status of information readout	Inspection	Observation	Displaying latest accurate reading	Externally measure parameters displayed on readout and compare with displayed values.	Unverified
1.3	Performance	Ability to identify tuning forks that are in/out of specification needed for successful device operation	Demonstration	Analysis	Devices do not fail due to tuning fork manufacturing defects	Use Six Sigma manufacturing techniques to analyze performance and failure root cause analysis	Unverified
1.4, 3.3	Performance	Status of operation in 14T magnetic field and 7mK environment	Test	Observation	Successful operation	Observe regular performance under the outlined condtions.	Unverified
1.5	Performance	Noise and EMI measurements	Analysis	Measurement	TBD at clients discretion	Measure noise and EMI of device during all operation modes.	Unverified
1.6, 4.1	Performance	Heat produced during operation	Analysis	Measurement	TBD at clients discretion	Measure heat/temperature change of device during all operation modes.	Unverified

5.2. Final Material Handoff & Deliverables

At the close of the project, the team will transfer the following physical assets and documentation to the client, Maybell Quantum.

Physical Assets:

- Prototype: readout device (readout device)
- Prototype: HTC (helium thermometry cell)
- Prototype: Mechanical apparatus to standardize tuning fork extraction from manufacturer's housing

Documents:

- Final Design Report
- Bill of Materials (BOM)

- CAD models, circuit schematics, and any other development documentation
- Manufacturing procedure document
- Test and calibration data and results

6. Management Plan

6.1. Team Members

Table 6 represents the team members highlighting their majors, skills, and roles.

Table 6: Team Members

Name	Role	Major	Skills	Email
Nathan Johnston	CFO (Chief Financial Officer)	Electrical Engineer	Digital and Analog design, Device Physics, Microwave theory	nathan_johnston@mines.edu
Zurin Trzaska	Scrum Master	Electrical Engineer	Signal Analysis, PCB (Printed Circuit Board) Design,	zurin_trzaska@mines.edu
Evan Christie	Communication lead	Mechanical Engineer, Physics minor	Materials physics, CAD (Computer-Aided Design), instrumentation	christie@mines.edu
Ryan Montoya		Mechanical Engineer	Fluid Mechanics, Manufacturing, Materials	ryan_montoya@mines.edu
Anastasiia Usmanova		Mechanical Engineer	Thermal fluids, CAD (Computer-Aided Design), Instrumentation	anastasiia_usmanova@mines.edu
Enzo Guglielmo		Electrical Engineer	RF Electronics, Electromagnetic compatibility	enzo_guglielmo@mines.edu



Figure 1: Picture of the team with the names of each member

6.2. Project Management Tools

The team will be using Scrum management framework. The team will use the MS Teams Planner tool for internal scheduling. The team created an outline for the future sprints and will be adding detailed tasks and assigning them as the workload is clear.

The team will use the MS Teams group within only the members for exchanging internal files, as well as a group with the client and Technical Advisors for exchanging external files. The client has confirmed that the MS Teams group is the preferred way of communication and file exchange.

6.3. Communication Strategy & Meeting Cadence

The team has reserved Hill Hall 378 from 8:00 am to 11:00 am, starting October 6th. The team will be meeting there when not impeded by scheduled course lectures.

In addition, the team has established weekly meetings on Tuesdays from 3:30 pm to 5:00 pm, during that time the team will be working on assignments. If there is not enough time to complete an assignment, team members are also available Mondays, Wednesdays, and Fridays from 5:00 pm to 6:00 pm and can meet during that time.

The team is planning to have the upcoming reports done at least 24 hours before the due date. The team will make sure everyone has a chance to review the document before submission.

Kyle Thompson, the founder of Maybell Quantum as well as the client, has delegated David Kendellen to be the primary touchpoint for this project. The team plans to meet with Kyle solely when questions that only he can answer arise. Kyle may attend the meetings whenever he is available.

6.4. Location of Work & Resources

The client's lab is located in Labriola, it is anticipated that most of the work can be completed there. The team may utilize additional resources such as physics labs as needed. Should the need present itself, the team is prepared to search for additional technical advisors and lab space.

References

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[3] D. I. M. C. S. N. F. e. a. Badley, "Thermometry in Normal Liquid 3He Using a Quartz Tuning Fork Viscometer," *Journal of Low Temperature Physics,* vol. 171, pp. 750-756, 2013.

[4] CIPM, "CIPM," 2000. [Online]. Available: https://www.bipm.org/documents/20126/41791796/PLTS-2000.pdf/.

[5] National Institute of Standards and Technology, "National Institute of Standards and Technology," 24 October 1989. [Online]. Available: https://www.nist.gov/system/files/documents/pml/div685/grp01/ITS-90_metrologia.pdf.

[6] ASTM International, "ASTM International," 14 May 2021. [Online]. Available: https://compass.astm.org/content-access?contentCode=ASTM%7CE0595-15R21%7Cen-US.

Appendix

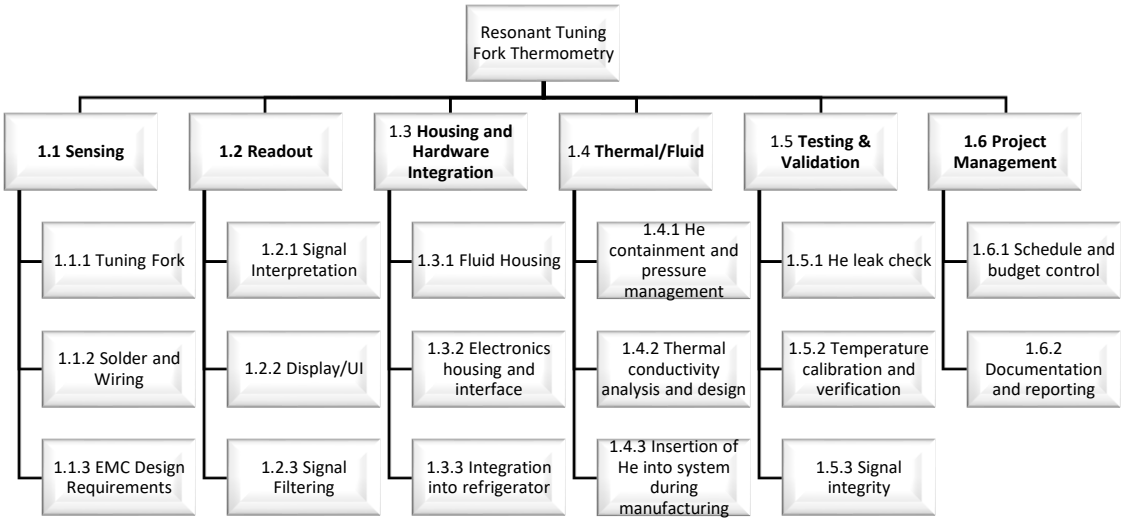
a. Table 3, section 3.2

Parent ID	Require	Requirement Type	Status	Requirement	Verification Type	Rationale/Justification
SubSys1	1.1	Performance	Draft	Measure temperature within +/- 1mK at low temperatures.	Test	Precision specified by client.
SubSys1	1.2	Performance	Draft	Measurement Lower Limit: Measure the temperature down to 7 mK	Test	Limit specified by client.
SubSys3	3.1	Functional	Draft	Contain ^{3}He/^{4}He mixture with no detectable leakage	Test	Any He leak will compromise function of device.
SubSys3	3.2	Functional	Draft	Demonstrate manufacturability of the design within unit cost budget, and	Demonstration	Client wants device to be manufacturable.
SubSys2	2.1	Performance	Draft	Measurement polling rate of less than	Test	Polling rate specified by
SubSys2	2.2	Functional	Draft	Device integrated on secondary equipment rack, using available power supplies and physical rack space	Inspection	Installation parameters specified by client.
SubSys2	2.3	Functional	Draft	Readout info: Display on rack with voltage, current, frequency, and calculated temperature	Inspection	Temperature display required by client, other values used for debugging and
SubSys1	1.3	Performance	Draft	Develop calibration scheme for fork-to-fork manufacturing variance	Demonstration	Contributes to manufacturability and consistent function of device
SubSys1, SubSys3	1.4, 3.3	Performance	Draft	Operational reliability in low temp and high magnetic fields up to 14T	Test	Operate in conditions and environment specified by
SubSys1	1.5	Performance	Draft	Minimize noise and EMI generation within the chamber	Analysis	Cannot interfere with performance of fridge or
SubSys1, SubSys4	1.6, 4.1	Performance	Draft	Minimize heat generation within chamber through wiring and non-magnetocaloric materials	Analysis	Cannot interfere with performance of fridge or devices within fridge.

b. Table 5, section 5.1

Requirement ID	Requirement Type	Performance Parameter	Verification Type	Measurement Method	Target/ Threshold (Acceptance Criteria)	Verification Procedure	Status
1.1	Performance	Difference between device measured temp and calibrated instrument measured temp	Test	Calculation	1mK in the low temperature range (<1K)	Compare device's temperature measurement to existing thermometry solution's temperature measurement.	Unverified
1.2	Performance	Difference between device measured temp and calibrated instrument measured temp	Test	Calculation	7mK with precision described by requirement 1.1	Compare device's temperature measurement to existing thermometry solution's temperature measurement.	Unverified
3.1	Functional	ppm/s He detected	Test	Measurement	Below detection limit of He leak check device	Use vacuum/helium leak detection device.	Unverified
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2.3	Functional	Status of information readout	Inspection	Observation	Displaying latest accurate reading	Externally measure parameters displayed on readout and compare with displayed values.	Unverified
1.3	Performance	Ability to identify tuning forks that are in/out of specification needed for successful device operation	Demonstration	Analysis	Devices do not fail due to tuning fork manufacturing defects	Use Six Sigma manufacturing techniques to analyze performance and failure root cause analysis	Unverified
1.4, 3.3	Performance	Status of operation in 14T magnetic field and 7mK environment	Test	Observation	Successful operation	Observe regular performance under the outlined conditions.	Unverified
1.5	Performance	Noise and EMI measurements	Analysis	Measurement	TBD at clients discretion	Measure noise and EMI of device during all operation modes.	Unverified
1.6, 4.1	Performance	Heat produced during operation	Analysis	Measurement	TBD at clients discretion	Measure heat/temperature change of device during all operation modes.	Unverified

c. Work Breakdown Structure



d. Gantt chart /project schedule

